Bo Li | 李波

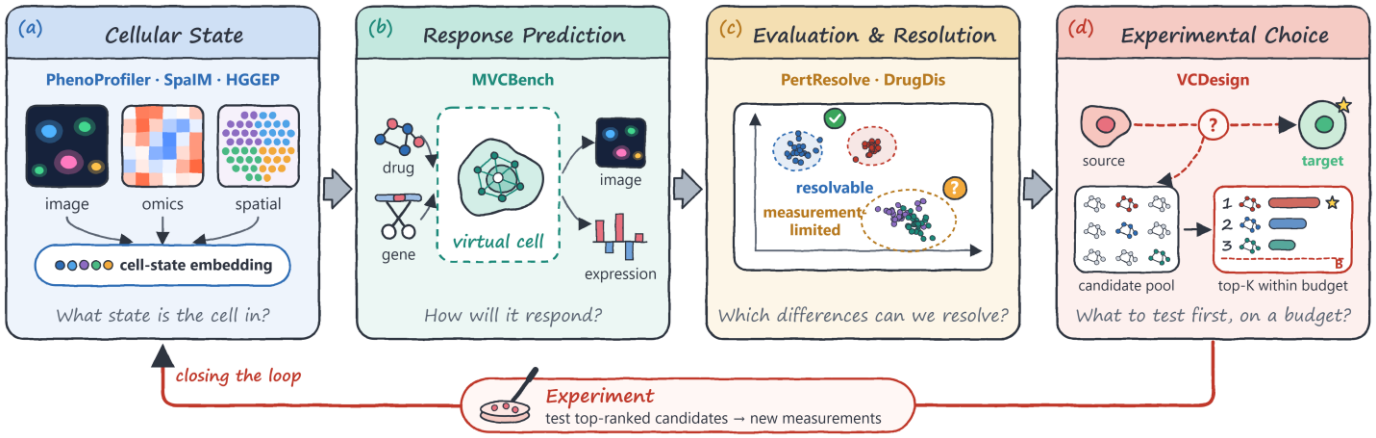
Ph.D. Candidate in Computer Science, University of Macau
Visiting Student, National University of Singapore

Multimodal Virtual Cells · **Perturbation Modeling** · **Intervention Design**

Email
Homepage
Google Scholar
GitHub
ORCID

RESEARCH PROFILE

I develop AI methods that turn multimodal cellular measurements into experimental decisions. My work spans representation learning across microscopy, single-cell and spatial omics (two first-author Nature Communications papers), measurement-aware evaluation of perturbation-response models, and finite-budget intervention design for virtual cells. The unifying questions are what experiments can reliably resolve, what models truly generalize across cellular contexts, and which interventions are worth testing next.



EDUCATION & APPOINTMENTS

University of Macau — Aug 2024 – Jun 2027 (expected)
Ph.D. in Computer Science | Advisor: Prof. Bob Zhang; Co-advisor: Prof. Qianqian Song

National University of Singapore — Jun 2026 – Jun 2027
Visiting Ph.D. Student | Host: Prof. Yang Zhang

Beijing University of Technology — Sep 2021 – Jul 2024
M.Eng. in Electronic Information | Advisors: Prof. Yong Zhang and Prof. Baocai Yin

Beijing Information Science & Technology University — Sep 2017 – Jul 2021
B.Eng. in Robotics Engineering | Advisor: Prof. Hongbo Huang

9	2	569	1,000+
First-author journal articles	First-author *Nature Communications*	citations · h-index 13 Google Scholar · Oct 2026	**GitHub stars** Across open-source projects

RESEARCH PROGRAM & SELECTED PUBLICATIONS

7 core works · † equal contribution

CELLULAR STATE PhenoProfiler · SpaIM · HGGEP

How can heterogeneous biological measurements reveal cellular phenotypes and states?

1. **PhenoProfiler: Advancing Phenotypic Learning for Image-based Drug Discovery**
   Bo Li, Bob Zhang, …, Mengran Li, Yong Zhang, Qianqian Song · *Nature Communications* 17, 793 (2026)
   Learns cellular representations from high-content microscopy for image-based compound profiling.
2. **SpaIM: Single-cell Spatial Transcriptomics Imputation via Style Transfer**
   Bo Li†, Ziyang Tang†, …, Baijian Yang, Jing Su, Qianqian Song · *Nature Communications* 16, 7861 (2025)
   Connects single-cell and spatial transcriptomics through style-transfer-based expression imputation.
3. **Gene Expression Prediction from Histology Images via Hypergraph Neural Networks**
   Bo Li, Yong Zhang, …, Mengran Li, Guangyu Wang, Qianqian Song · *Briefings in Bioinformatics* 25(6), bbae500 (2024)
   Uses hypergraph learning to capture higher-order structure for histology-to-gene-expression prediction.

RESPONSE PREDICTION **MVCBench**

What information transfers across perturbations, modalities, and cellular contexts?

4. **MVCBench: A Multimodal Benchmark for Drug-induced Virtual Cell Phenotypes**
   Bo Li†, Qing Wang†, Shihang Wang†, …, Xiaojun Yao, Chuxia Deng, Qianqian Song · *bioRxiv* (2026)
   Benchmarks multimodal perturbation prediction under compound, cell-line, plate, and dataset shifts.

EVALUATION & RESOLUTION **PertResolve · DrugDis**

When does predictive performance reflect reproducible, biologically useful response information?

5. **Measurement resolution constrains fine-grained perturbation prediction**
   Bo Li†, Chengyang Zhang†, Mengran Li†, …, Yuhao Yi, Jiancheng Lv, Yang Zhang · *Under review* (2026)
   Separates model limitations from measurement-resolution limits by testing which fine-grained perturbation distinctions experiments can reproducibly resolve.

6. **Disentangling general and context-specific effects in drug-response prediction**
   Bo Li†, Chengliang Liu†, Yuzhong Peng†, …, Shenghui Huang, Chuxia Deng, Yang Zhang · *Under review* (2026)
   Separates general drug and sample effects from drug–sample interactions, revealing when aggregate accuracy masks weak recovery of context-specific response structure.

EXPERIMENTAL CHOICE **VCDesign**

Which interventions are worth testing under a finite experimental budget?

7. **VCDesign: Finite-Budget Intervention Design for Virtual Cells**
   Bo Li†, Lin Wang†, Bob Zhang, …, Chengliang Liu, Zhiyuan Liu, Yang Zhang · *Under review* (2026)
   Formulates finite-budget intervention ranking, evaluates selected candidates against independently measured held-out outcomes, and transfers historical evidence to response-unseen candidates.

SELECTED OTHER FIRST-AUTHOR PUBLICATIONS

full list of 39 works on Google Scholar

8. **Multi-scale Hypergraph-based Feature Alignment Network for Cell Localization**
   Bo Li, Yong Zhang, Chengyang Zhang, Xinglin Piao, Yongli Hu, Baocai Yin · *Pattern Recognition* 149, 110260 (2024)
9. **Exponential Distance Transform Maps for Cell Localization**
   Bo Li, Jie Chen, Hang Yi, …, Yongquan Yang, Qikui Zhu, Hong Bu · *Eng. Appl. Artif. Intell.* 132, 107948 (2024)
10. **Lite-UNet: A Lightweight and Efficient Network for Cell Localization**
    Bo Li, Yong Zhang, Yunhan Ren, Chengyang Zhang, Baocai Yin · *Eng. Appl. Artif. Intell.* 129, 107634 (2024)

Co-corresponding-author manuscript

11. **When do single-cell response distributions improve drug retrieval?**
    Huihong Wang, Xiaoping Liu, Chen Chen, Qiuyan Sun#, **Bo Li#**, Bob Zhang# · *Under review* (2026)

OPEN SOURCE & RESEARCH SYSTEMS

1,000+ GitHub stars in total

Nature-Paper-Skills	Agent skills for scientific writing, revision, auditing and resubmission.	**583 stars**
Awesome Virtual Cell	Research landscape: 328 papers · 20 themes · interactive map.	**324 stars**
DrugClaw	Agentic RAG for drug research · 57 skills · 15 task categories.	**116 stars**

INVITED TALKS & MENTORING

Invited talk: PhenoProfiler, Prof. Guangyu Wang's group, Weill Cornell Medicine & Houston Methodist — Feb 2025

Mentoring: six student-led papers from co-mentored BJUT graduate students (New Phytologist, IEEE JBHI, ACM TOMM, CVIU).

HONORS, AWARDS & PATENTS

National Scholarship	2022	**Beijing Outstanding Graduate**	2021, 2024
Top 100 Graduates, BJUT	2024	**Xiaomi Scholarship**	2023
First-Class Academic Scholarship, BJUT	2022, 2023	**Ph.D. Full Scholarship**, UM	2024
First Prize, S&T Innovation Scholarship, BISTU	2021		

Patents: two granted Chinese invention patents on cell localization, co-inventor (2026): CN115457547B · CN115810046B

ACADEMIC SERVICE

Reviewer: ICLR 2027 (conference); Science Advances; IEEE TPAMI, TIP, TNNLS; Medical Image Analysis; Bioinformatics; Briefings in Bioinformatics; Pattern Recognition; BMC Biology; EAAI; Expert Systems with Applications (journals).